studies focus on reducing hallucination as an undesirable effect, while creativity research rarely considers the potential role of hallucination in generating innovative content.

Therefore, at present, there is no systematic study investigating the relationship between hallucination and creativity in LLMs. This work aims to bridge this research gap.

3 Methodology

In this study, we propose a three-stage evaluation framework *HCL (Hallucination-Creativity Layer-wise)* to explore the relationship between creativity and hallucination in LLMs layer-wise generations (Figure 2). First, to ensure the layer-wise output is generally meaningful, we obtain the responses sampled from each layer of LLM by leveraging the early-exit strategy (Section 3.1). Second, we propose the creativity metric and assign each response with both creativity and hallucination metrics (Section 3.2). Lastly, we propose the HCB score which will be used to optimize the trade-off between these creativity and hallucination metrics (Section 3.3).

3.1 Layer-wise Response Sampling

Unlike conventional decoding strategies that rely on the final layer's outputs, our key insight lies in *analyzing and potentially utilizing the responses from intermediate layers*. This design is based on the following key observations and findings:

- **Confidence is lower in earlier layers, enabling more diverse outputs.** During the decoding process of LLMs, earlier layers tend to exhibit higher uncertainty, preserving more possibilities in the generation process, as shown in Figure 3. This uncertainty allows them to produce more diverse and creative outputs. Furthermore, if these earlier layers can generate creative content with minimal impact on accuracy, it becomes feasible to directly extract responses from them improve the inference efficiency.

  The need for early exit. Since deeper layers tend to produce more conservative outputs, while some intermediate layers may already achieve an optimal balance between creativity and hallucination, terminating decoding at these layers can not only reduce computational overhead but also prevent creativity loss (Chuang et al., 2023).

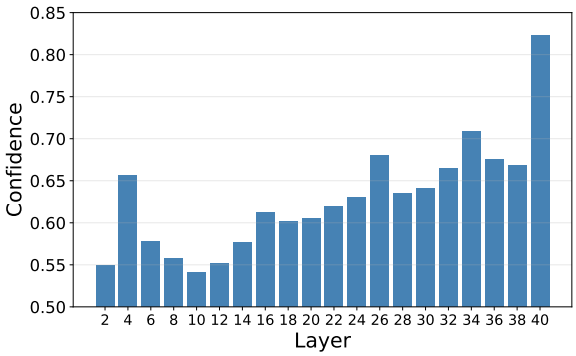

Figure 3: Confidence variations across layers in LLaMA2-13B. We adopt P(True) to allow each layer of the LLM to self-evaluate the average confidence among the corresponding sampled responses.

Based on these observations and assumptions, we aim to **analyze creativity and hallucination layer by layer** to achieve two objectives: (1) Conduct a more fine-grained investigation into their interaction during the response generation process of LLMs, unveiling their underlying mechanisms. (2) Identify the optimal decoding layer that allows the model to exit early while maintaining a favorable balance between creativity and factual accuracy, thereby reducing computational cost.

In order to better understand how creativity and hallucination evolve across different depths, we adopt a *Layer-Skip* strategy inspired by speculative decoding (Elhoushi et al., 2024). Specifically, given an input consisting of a question q and a shared prompt p, we sample responses generated from the earlier layers $\{\ell_1, \ell_2, \ldots, \ell_{N-1}\}$ (using speculative decoding) and the final layer ℓ_N (using standard autoregressive decoding) of the LLM. We denote the resulting response list as r, formally expressed as:

$$r = \{[r_1, r_2, \ldots, r_{N-1}], r_N\},$$
$$\text{where } r_i = \bigcup_{j=1}^{D} LLM_i^{(j)}(p(q)),\ i \in \{1, \ldots, N\}. \tag{1}$$

where i refers to the i-th layer of the LLM and D denotes the sampling times. Building upon the above procedure, we assigned $N \times D$ responses generated by each layer of the LLM to each question for subsequent layer-wise evaluation of the two metrics, creativity and hallucination.

3.2 Evaluation Metric

Hallucination. Following (Orgad et al., 2024), we define hallucination as any type of error gener-

ated by an LLM in our study. Hence, we have to justify the correctness of the responses generated by each decoding layer from LLM before evaluating their hallucination metrics. We adopt the following criteria for judging the correctness of free-form responses: if the generated response contains the correct answer, it is deemed correct; otherwise deemed hallucination. Based on the above, the hallucination metric of sampled layer-wise responses can be defined as follows,

$$\mathbf{S_H^i} = \frac{N_e^{(i)}}{D^{(i)}}, \quad \text{where } i \in \{1, \ldots, N\}. \quad (2)$$

where $N_e^{(i)}$ denotes the incorrect times and $D^{(i)}$ refers to the sampling time at *layer* i.

Creativity. Following the definition that creativity is both novel and useful (Jiang et al., 2024), we define the diversity of correct outputs as creativity. Therefore, when we filter out incorrect responses from the n responses, we need to group the semantically equivalent (Ribeiro et al., 2018) correct responses. To meet this requirement, we utilize a SentenceTransformer-based encoder, the pre-trained *all-MiniLM-L6-v2* model (Vergou et al., 2023), to extract dense semantic embeddings and group them as different semantic clusters based on the semantic-level similarity. Subsequently, we categorize the outputs based on group types and evaluate the creativity metric.

$$\mathbf{S_C^i} = N_D^{(i)}, \quad \text{where } i \in \{1, \ldots, N\}. \quad (3)$$

where $N_D^{(i)}$ is the count of unique semantic clusters at *layer* i.

3.3 HCB Calculation

Once we obtain the scores for **creativity** and **hallucination**, we need to evaluate the performance of each model layer in generation tasks. To achieve this, we propose a *Hallucination and Creativity Balanced (HCB)* Score, which combines **creativity** and **hallucination** using distinct normalization methods. Specifically, **creativity** is normalized via min-max scaling, while **hallucination** is quantified directly through the error rate. This score provides a unified metric to assess the model's ability to generate outputs that are both accurate and diverse, ensuring a balanced trade-off between creativity and hallucination.

We compute the HCB score S_{HCB}^iin the layer i as follows:

$$S_{HCB}^i = w_c \times S_C^i \ + \ w_h \times \left(1 - S_H^i\right),$$

where w_c and w_h are the weights corresponding to creativity and hallucination, respectively. Here, $w_c + w_h = 1$. Note that S_C^i is the normalized score, where S_C^i is the normalized creativity score, and S_H^i is the hallucination score, and S_{HCB}^i is the HCB score.

4 Experiments

In this section, we present the experimental setup, models, datasets, and discuss the key findings. Based on previous methods (Section 3.3), in all experiments, for each query, LLMs respond 50 times using the same prompt to ensure we have sufficient responses to evaluate the creativity and hallucination of LLMs.

4.1 Experimental Setups

Models We use four popular open-weight models: LLaMA 3.2-1B, LLaMA 2-7B, LLaMA 3-8B, and LLaMA 2-13B (Touvron et al., 2023). These models allow us to systematically analyze how model size and different layers influence the trade-off between creativity and hallucination.

Datasets For our experiments, we utilized two open-domain question answering (QA) datasets: TriviaQA(Joshi et al., 2017) and Natural Questions (NQ)(Kwiatkowski et al., 2019). These datasets are widely used in QA research, covering a vast range of real-world questions with multiple valid answers. They provide a suitable benchmark for evaluating LLMs in terms of information retrieval, factual generation, and creative expression.

TriviaQA: TriviaQA is a general knowledge QA dataset that spans multiple domains, including history, science, literature, sports, and entertainment. One of its key characteristics is that each question typically has multiple acceptable correct answers. This feature makes it ideal for assessing both the accuracy and creativity of LLMs, allowing evaluation even when models generate different but reasonable responses.

Natural Questions (NQ): Natural Questions, released by Google, consists of real user queries from Google Search, with answers typically extracted from Wikipedia, emphasizing factual consistency. In the latest version of the dataset, Natural Questions have evolved from multiple-choice to open-ended text generation, introducing more flexibility.

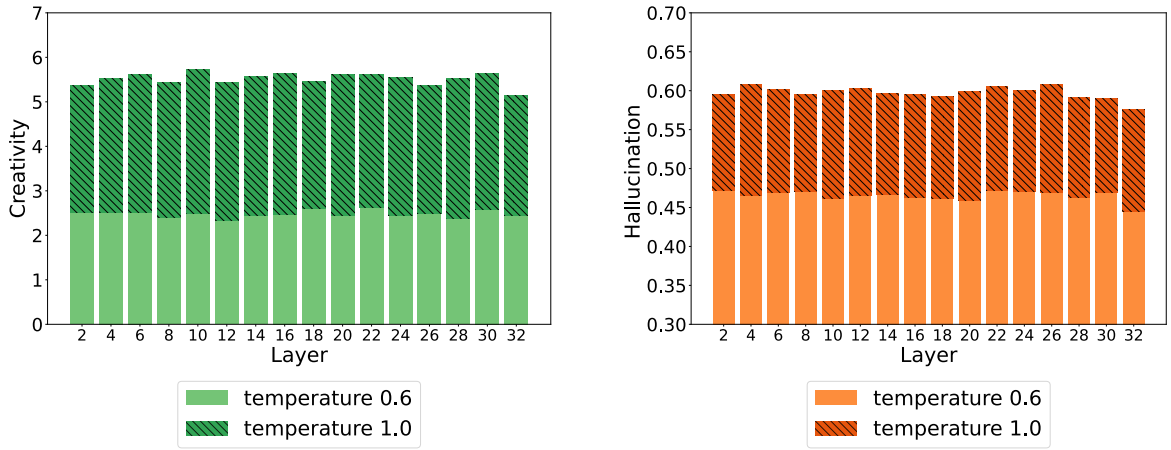

Figure 4: The variation of layer-wise creativity and hallucination metrics of the LLaMA3-8B when its temperature coefficient increases from 0.6 to 1.0 on TriviaQA benchmark.

Moreover, the dataset now include many questions with multiple valid answers, making it more suitable for assessing response diversity.

In this study, we specifically filtered questions with three or more correct answers to ensure sufficient answer diversity. This approach allows us to assess whether models can maintain factual accuracy while exhibiting creativity, providing a more comprehensive evaluation of LLM performance in open-domain QA tasks.

4.2 Explore the relationship between creativity and hallucination

In this part, we focus on analyzing the creativity and hallucination metrics of LLMs at each layer during response generation. Our experimental results reveal some fundamental relationships between the two dimensions, providing deeper insights into their interplay.

Creativity comes with hallucination. Existing studies mainly consider increasing the model temperature to enhance the diversity of LLM's generations, since the temperature parameter determines the smoothness of the probabilities while sampling and a higher temperature value indicates more diverse sampling(Peeperkorn et al., 2024). However, as the temperature parameter increases, both creativity and hallucination rates rise in a proportional manner, as shown in Figure 4. This suggests that higher temperature values encourage more diverse and novel outputs, fostering greater creativity by allowing the model to explore unconventional ideas. However, this exploratory behavior comes at a cost: an increased likelihood of generating factually inaccurate or unverifiable content.

This trade-off highlights the inherent tension between diversity-driven creativity and factual precision in LLMs. When the model is set to lower temperatures, it tends to produce more deterministic and factually consistent responses, but at the expense of originality and expressiveness. Conversely, when the temperature is raised, the model exhibits a greater degree of unpredictability, leading to more imaginative but less reliable outputs.

These findings align with previous studies, suggesting that LLMs' tendency to hallucinate is not just a flaw but a natural consequence of their generative flexibility.

Stronger models are more creative, though also more prone to hallucination. A second key observation from our experiments is that LLMs tend to exhibit higher levels of both creativity and hallucination. Specifically, model size appears to correlate positively with the generation of novel yet sometimes factually incorrect responses. For instance, smaller models such as LLaMA-3.2-1B tend to be more conservative in their outputs, often adhering closely to more predictable, template-like responses. While this makes them less prone to hallucination, it also limits their ability to produce highly original and imaginative content. In contrast, larger models (e.g., LLaMA-3-8B or LLaMA-13B) demonstrate a greater ability to generate complex and creative responses, but they are also more susceptible to producing hallucination(Figure 5). This suggests an intrinsic trade-off between model capacity and output reliability: as models become more expressive and generative, they also gain